CAST-Norm: Coupled Adaptive Spatio-Temporal Normalization for Multivariate Time Series Forecasting

Haihua Xu
yc37901@um.edu.mo
SKL-IOTSC, Department of CIS,
University of Macau
Macau SAR, China

Ziyue Peng
dc22951@um.edu.mo
Department of Mathematics,
University of Macau
Macau SAR, China

Runchang Liang
liangrc666@nenu.edu.cn
School of Information Science and Technology,
Northeast Normal University
Changchun, Jilin, China

Pengyang Wang*
pywang@um.edu.mo
SKL-IOTSC, Department of CIS,
University of Macau
Macau SAR, China

Abstract

Distribution shift fundamentally hinders the efficacy of deep models in Multivariate Time Series Forecasting (MTSF). While mitigating non-stationarity requires characterizing both intra- and inter-series shifts, existing literature is predominantly constrained by a limiting **"Decoupled View"**, erroneously treating them as orthogonal components. Specifically, proxy-based methods exhibit a "Spatial Stationarity Bias," while proxy-free methods reinforce decoupling via disjoint modules. We theoretically prove that these shifts are **mutually coupled**; consequently, addressing them in isolation neglects their inherent interaction. To bridge this gap, we propose **C**oupled **A**daptive **S**patio-**T**emporal **Norm**alization (CAST-Norm). Technically, CAST-Norm employs a *Hybrid Inter-Intra Series Normalization* strategy to isolate systematic shifts and yield spatiotemporally stationary patterns. Furthermore, to overcome the "Fallacy of Independent Restoration," we design a *Coupled Adaptive Denormalization* mechanism. Instead of reversing statistics independently, it leverages learned spatiotemporal dependencies to jointly modulate the restoration process, ensuring the recovered distribution accurately reflects the non-stationary future. Extensive experiments on eight real-world datasets across four backbone architectures demonstrate that CAST-Norm consistently outperforms state-of-the-art methods. Code is available at https://github.com/xhhmacau/CAST_Norm.

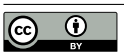
CCS Concepts

• **Mathematics of computing** → *Time series analysis*; • **Information systems** → **Data stream mining**.

Keywords

Time Series Forecasting; Distribution Shift; Normalization

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License.
KDD '26, Jeju Island, Republic of Korea
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2259-2/2026/08
https://doi.org/10.1145/3770855.3817667

ACM Reference Format:
Haihua Xu, Ziyue Peng, Runchang Liang, and Pengyang Wang. 2026. CAST-Norm: Coupled Adaptive Spatio-Temporal Normalization for Multivariate Time Series Forecasting. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3770855.3817667

1 Introduction

Multivariate Time Series Forecasting (MTSF) is foundational to critical applications ranging from traffic management [10] and energy planning [36] to weather forecasting [14] and financial analysis [19]. However, its real-world efficacy is persistently hindered by distribution shift—where data statistics evolve over time, rendering historical patterns insufficient for future prediction [7, 17]. Effectively addressing non-stationarity in these systems requires characterization of two distinct dimensions of shift: *intra-series shift* (temporal variations within individual variables) and *inter-series shift* (dynamic evolution of correlations among variables) [12, 27].

Despite extensive efforts, we observe that existing shift-mitigation methods are predominantly constrained by a limiting **"Decoupled View."** This perspective treats intra-series and inter-series shifts as orthogonal, independent components to be addressed in isolation. Specifically, current methodologies fall into two paradigms, both trapped by this decoupled assumption: 1) Proxy-based methods, exemplified by normalization approaches [9, 13], function as model-agnostic plug-ins mapping non-stationary data onto stable distributions. However, these approaches suffer from a "Spatial Stationarity Bias": by prioritizing temporal normalization, they implicitly assume spatial correlations remain static [18] or are recoverable via historical scaling [6], thereby treating spatial dynamics as decoupled from temporal fluctuations. 2) Proxy-free methods design end-to-end architectures to decompose raw data into invariant and variant patterns [27]. While acknowledging both shift dimensions, these architectures typically isolate them via disjoint, parallel modules. This design explicitly embodies the decoupled view, operating on the assumption that spatial dependencies and temporal trends are orthogonal and can be extracted in isolation.

We argue that this decoupled perspective is often suboptimal in highly dynamic contexts because intra- and inter-series shifts are

mutually coupled. Practically, this means temporal shifts within individual variates and the evolution of inter-variate dependencies are not isolated, but jointly drive distribution changes. As mathematically demonstrated in Appendix A.1, coupling terms in the error bound indicate that dynamic variations in one dimension inherently constrain the other. Addressing these shifts in isolation neglects their interaction, leading to compromised stationarity and inaccurate recovery. *To operationalize this coupled view, we aim to synthesize the strengths of both paradigms: inheriting the model-agnostic universality of normalization-based approaches while integrating the explicit spatial modeling of inter-series dynamics from proxy-free architectures, ultimately resolving their shared decoupled bottleneck within a unified framework.* However, elevating the normalization paradigm to explicitly model spatiotemporal entanglement presents two interrelated challenges:

- ***Challenge 1: The Dilemma of Spatiotemporal Stationarity Construction and Dynamic Shift Extraction.*** Normalization faces a non-trivial dual objective: simultaneously constructing strictly stationary inputs for the backbone model and extracting accurate shift information for future restoration. First, regarding *stationarity construction*, the inherent entanglement of inter- and intra-series dynamics renders sequential removal prone to mutual interference, where stabilizing one dimension inadvertently destabilizes the other. Second, regarding *shift extraction*, raw inter-series shifts are often noisy mixtures of *learnable systematic evolution* (persistent changes driven by latent factors) and *unlearnable stochastic perturbations* (transient, idiosyncratic noise) [5, 20]. Since only the systematic component is predictive, using raw, noise-contaminated shifts for restoration inevitably injects bias. The core difficulty, therefore, lies in designing a mechanism that ensures rigorous spatiotemporal stationarity without interference, while simultaneously distilling systematic evolution from stochastic perturbations.
- ***Challenge 2: The Fallacy of Independent Restoration.*** Standard methods typically restore the original scale by simply reversing historical normalization statistics [13]. However, accurate denormalization requires predicting the unknown distribution of the future [9]. Since spatial and temporal shifts are mutually dependent, restoring them independently ignores the system's joint evolution. Predictive accuracy thus necessitates an integrated mechanism that leverages learned dependencies to adaptively modulate future distribution restoration.

To bridge this gap, we propose **C**oupled **A**daptive **S**patio-**T**emporal **Norm**alization (CAST-Norm), a framework that advances beyond decoupled processing by introducing a practical mechanism for holistic modeling of distribution shifts. In the normalization phase, we introduce a *Hybrid Inter-Intra Series Normalization* strategy. We identify a critical ordering principle: performing spatial normalization prior to temporal normalization. This is based on the fact that spatial dependencies, commonly quantified by metrics such as Pearson correlation [27], remain invariant under scale changes, ensuring rigorous stationarity without interference (proven in Appendix A.2). To extract the systematic shift component, we propose a *Low-rank Manifold Spatial Decoupling* module grounded in the manifold hypothesis, which filters stochastic perturbations to isolate genuine spatial evolution. In the denormalization phase, we design a *Coupled Adaptive Denormalization* module. Instead of independently reversing shifts, we learn spatiotemporal coupling patterns to jointly modulate the restoration, ensuring the recovered distribution accurately reflects the non-stationary future.

Our contributions are summarized as follows:

- **Theoretical Insight & Coupled Perspective**: We identify the **"Decoupled View"**—a pervasive but limiting assumption in shift-mitigation methods. By mathematically formalizing the interaction between intra- and inter-series shifts, we highlight the practical benefits of joint spatiotemporal modeling over strictly decoupled approaches.
- **Innovative Framework (CAST-Norm):** We propose CAST-Norm, a unified framework that addresses coupled distribution shifts by integrating *Hybrid Inter-Intra Series Normalization* for rigorous stationarity construction and *Coupled Adaptive Denormalization* for holistic distribution restoration.
- **Comprehensive Evaluation:** We conduct extensive experiments on eight real-world datasets across four backbone architectures. The results demonstrate that CAST-Norm consistently outperforms state-of-the-art normalization-based methods, validating the necessity of modeling spatiotemporal coupling.

2 Related Work

2.1 Distribution Shift in Time Series

Distribution shift refers to the phenomenon where the data distribution evolves over time, posing a fundamental challenge to the generalization capability of forecasting models [8]. In Multivariate Time Series Forecasting, this issue is categorized into two dimensions: *intra-series shift* (temporal variations in individual variates) and *inter-series shift* (dynamic evolution of correlations among variables) [12, 27]. To mitigate such shifts, normalization-based methods [9, 13, 30] predominantly focus on neutralizing intra-series shift via reversible transformations. However, these approaches typically treat variables as independent entities, operating under a spatially agnostic assumption that fails to capture the dynamic shifts in inter-series correlations. Even advanced frameworks [16, 27, 28] that incorporate spatial dynamics inadvertently adhere to a "Decoupled View." These methods assume that intra- and inter-series shifts are orthogonal components that can be disentangled and modeled in isolation. Crucially, this overlooks the *reciprocal interaction* where temporal non-stationarity and spatial dynamic correlations mutually evolve and constrain one another. Conversely, we theoretically validate this inherent *spatiotemporal coupling* and explicitly model the entanglement to ensure holistic distribution recovery.

2.2 Normalization-based Methods for MTSF

The normalization-denormalization framework [13] has become a dominant strategy for handling distribution shift, though it originally relied on reversing static historical statistics for restoration. Recognizing that future distributions inherently drift from the past, subsequent approaches [9, 18, 30] argue that accurate recovery necessitates predicting dynamic statistics rather than simply reusing static parameters. For instance, Dish-TS [9] decouples normalization statistics via a Dual-CONET framework, learning distinct coefficients to capture temporal shifts. Recent advancements have

further refined intra-series modeling: SAN [18] introduces slice-based normalization for local statistics, while FAN [30] extends this to the spectral domain. However, these methods typically exhibit a "Spatial Stationarity Bias." ST-Norm [6] attempts to incorporate spatial information but defines it merely as static scale disparities rather than dynamic correlations. Similarly, CAN-ST [29] relies on discrete, static pattern matching, lacking the capacity to model continuous interactions. Consequently, these frameworks remain constrained by the "Decoupled View," implicitly assuming that spatial correlations remain static [9, 13, 18] or are recoverable via historical scaling [6]. In contrast, our framework advances this paradigm by explicitly modeling the spatiotemporal coupling to overcome these static assumptions.

3 Problem Formulation

In multivariate time series forecasting, we aim to predict future values based on historical observations. Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T] \in \mathbb{R}^{N \times T}$ denote a multivariate time series consisting of N variates over T timestamps, where $\mathbf{x}_t \in \mathbb{R}^N$ represents the value vector at time step t. Given a lookback window of length L, the historical input sequence is denoted as $\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{N \times L}$. The objective is to forecast the future sequence over a horizon H, denoted as the ground truth $\mathbf{X}_{t+1:t+H} \in \mathbb{R}^{N \times H}$. To this end, we aim to learn a mapping function $\mathcal{F}_{\boldsymbol{\theta}} : \mathbb{R}^{N \times L} \mapsto \mathbb{R}^{N \times H}$ parameterized by $\boldsymbol{\theta}$ to generate the prediction $\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{F}_{\boldsymbol{\theta}}(\mathbf{X}_{t-L+1:t})$. The parameters $\boldsymbol{\theta}$ are optimized by minimizing the Mean Squared Error (MSE) between the predictions and the ground truth:

$$\mathcal{L} = \frac{1}{H} \left\| \hat{\mathbf{Y}}_{t+1:t+H} - \mathbf{X}_{t+1:t+H} \right\|^2. \tag{1}$$

4 Methodology

4.1 Overview

We propose **Coupled Adaptive Spatio-Temporal Normalization (CAST-Norm)**, a unified framework designed to transcend the limitations of the "Decoupled View." Our pipeline is explicitly structured into two phases (Figure 1): Normalization (HISN) to systematically remove shifts, and Denormalization (CAD) for coupling-aware restoration. Phase 1 initiates with the *Hybrid Inter-Intra Series Normalization (HISN)* module, enforcing a strict spatial-then-temporal ordering (validated in Appendix A.2). Here, spatial normalization first separates stable patterns from shifts, and the *Low-rank Manifold Spatial Decoupling (LMSD)* filters out high-rank stochastic noise to extract the pure systematic inter-series shift; subsequently, temporal normalization processes the remaining invariant patterns to yield rigorously stationary inputs for the backbone. In Phase 2, the *Coupled Adaptive Denormalization (CAD)* module addresses the fact that restoring spatial and temporal shifts independently inherently ignores their joint system evolution. Thus, we first deploy the *Spatio-Temporal Coupling Perception (STCP)* module to compute dynamic interaction contexts, which subsequently guide the *Coupling-Aware Spatio-Temporal Denormalization (CASD)* to adaptively recalibrate distribution statistics via a coupling-gated fusion mechanism, ensuring precise restoration.

4.2 Hybrid Inter-Intra Series Normalization

In this module, distinct from prior normalization-based methods that focus exclusively on intra-series shift, we explicitly account for inter-series shift. Consequently, the primary objective of this module is to generate series representations that exhibit stationarity across both temporal and spatial dimensions. Meanwhile, we capture both inter-series and intra-series shifts, which serve as the essential basis for the subsequent denormalization phase.

4.2.1 ***Low-rank Manifold Spatial Decoupling.*** To simultaneously extract spatially invariant representations and isolate purified inter-series shifts, we propose the *Low-rank Manifold Spatial Decoupling (LMSD)* method. Grounded in the Manifold Hypothesis, this module filters stochastic noise to preserve the systematic evolution of spatial dynamics for effective denormalization.

Momentum-based Topology Memory. To establish a robust reference for spatial dependencies, we construct a long-term topology memory. Given the instantaneous correlation matrix $\mathbf{A}_t = \mathrm{Corr}(\mathbf{X}_{t-L+1:t})$, we maintain a stable memory matrix $\mathbf{M}$ updated via an Exponential Moving Average (EMA) strategy [22]:

$$\mathbf{M}^{t+1} \leftarrow \alpha \mathbf{M}^t + (1-\alpha)\mathbf{A}_t, \tag{2}$$

where $\alpha \in (0,1)$ controls the update momentum. This mechanism accumulates historical interactions into $\mathbf{M}$, explicitly capturing stable, long-term systematic patterns.

Invariant Pattern Extraction. We first disentangle the stationary structure from the raw data. To this end, we employ a learnable spatial mapping, denoted as $\mathcal{P}$, to map the input series $\mathbf{X}_{t-L+1:t}$ into the invariant patterns $\mathbf{X}_{\mathrm{inv},t}$:

$$\mathbf{X}_{\mathrm{inv},t} = \mathcal{P}(\mathbf{X}_{t-L+1:t}), \tag{3}$$

where the projection mapping $\mathcal{P}(\cdot)$ is implemented as a Multi-Layer Perceptron. To ensure structural consistency, we impose a topological constraint aligning the correlation of $\mathbf{X}_{\mathrm{inv},t}$ with the stable memory $\mathbf{M}^t$:

$$\mathcal{L}_{\mathrm{stable}} = \left\| \mathrm{Corr}(\mathbf{X}_{\mathrm{inv},t}) - \mathrm{StopGrad}(\mathbf{M}^t) \right\|^2. \tag{4}$$

Low-rank Variant Pattern Purification. After extraction, we obtain the raw residual $\mathbf{X}_{\mathrm{diff},t} = \mathbf{X}_{t-L+1:t} - \mathbf{X}_{\mathrm{inv},t}$. However, treating this residual directly as the inter-series shift is problematic due to substantial noise contamination. To address this, we establish a theoretical framework grounded in the *Manifold Hypothesis* to guide our purification design.

1) Theoretical Foundation: The Manifold Hypothesis posits that high-dimensional real-world data is not uniformly distributed but concentrates near low-dimensional geometric structures determined by latent governing laws [2]. In the context of MTSF, established theoretical frameworks suggest that while stochastic noise fills the ambient high-dimensional space, genuine inter-series dependencies evolve along a low-dimensional manifold [23, 32]. Formally, we model $\mathbf{X}_{\mathrm{diff},t}$ under the following assumption:

[Signal-Noise Decomposition] Given a valid community partition, the residual decomposes into $\mathbf{X}_{\mathrm{diff},t} = \mathbf{X}_{\mathrm{va},t} + \mathbf{N}_t$:

- **Systematic Evolution ($\mathbf{X}_{\mathbf{va},t}$):** A **low-rank** component representing coherent shifts driven by latent factors shared within communities ($\mathrm{rank}(\mathbf{X}_{\mathrm{va},t}) \le K \ll N$).

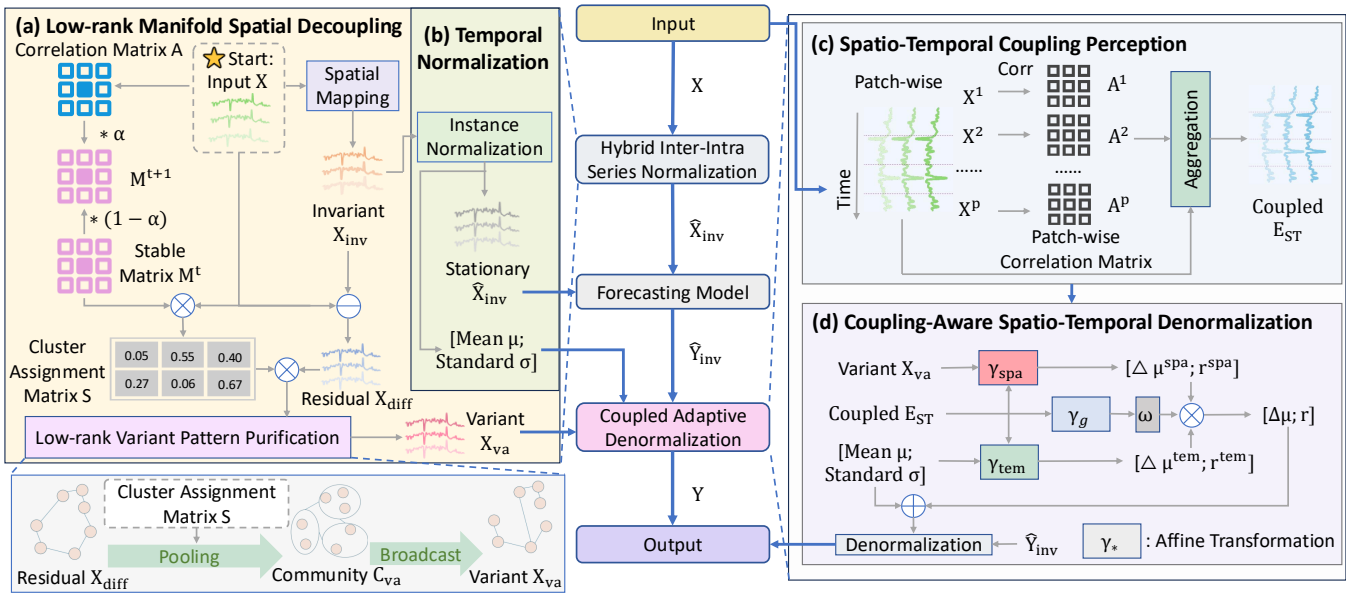


Figure 1: An illustration of the proposed CAST-Norm framework. The pipeline comprises four integrated components: (a) The *Low-rank Manifold Spatial Decoupling* module (§4.2.1) that spatially decouples the input series to obtain invariant patterns and purified variant patterns via manifold projection. (b) The *Temporal Normalization* module (§4.2.2) that eliminates intra-series variations to yield rigorously stationary inputs for the backbone forecaster. (c) The *Spatio-Temporal Coupling Perception* module (§4.3.1) that extracts dynamic coupling contexts through patch-wise graph aggregation. (d) The *Coupling-Aware Spatio-Temporal Denormalization* module (§4.3.2) that adaptively restores the distribution using context-aware affine parameters conditioned on the learned coupling patterns.

- **Stochastic Perturbations ($\mathbf{N}_t$):** A **high-rank** component representing idiosyncratic, spatially uncorrelated noise that is orthogonal to the community structure.

Based on this, we propose to recover systematic evolution (variant pattern) $\mathbf{X}_{\text{va},t}$ via a rank-constrained operator.

Proposition 4.1 (Rank-Constrained Noise Attenuation). *Let $\mathcal{P}_{\mathcal{S}}$ be a projection operator onto the subspace spanned by the stable community structure* $\mathbf{S}$*. This operator acts as a structural filter: it preserves the coherent energy of systematic shifts while attenuating the magnitude of stochastic perturbations by a factor proportional to the community size.*

Proof Sketch. The operator $\mathcal{P}_{\mathcal{S}}$ enforces an information bottleneck by constraining the output to a rank-K subspace. For systematic shifts $\mathbf{X}_{\text{va},t}$ aligned with $\mathbf{S}$, the projection is lossless. Conversely, for an incoherent noise impulse in $\mathbf{N}_t$, the projection distributes its energy across the entire community, effectively diluting the noise magnitude by $1/M_k$ (where M_k is the community size). See Appendix A.3 for the rigorous derivation and analysis. □

2) Differentiable Community Assignment: Guided by Assumption 1, establishing the correct low-rank basis $\mathbf{S}$ is a prerequisite. To ensure the identified communities reflect stable long-term dependencies rather than being distorted by stochastic perturbations, we derive the partition from the historical stable memory $\mathbf{M}^t$ instead of the noisy instantaneous adjacency. We first calculate the normalized stable adjacency matrix via $\hat{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}}(\mathbf{M}^t + \mathbf{I})\mathbf{D}^{-\frac{1}{2}}$, where $\mathbf{I}$ is the identity matrix representing self-loops, and $\mathbf{D}$ denotes the degree matrix of $(\mathbf{M}^t + \mathbf{I})$. Subsequently, a GCN-based encoder processes this stable graph to generate node embeddings, which are projected to obtain the soft cluster assignment matrix $\mathbf{S}_t \in \mathbb{R}^{N\times K}$:

$$\mathbf{H}_t = \text{ReLU}\big(\hat{\mathbf{A}}\mathbf{X}_{t-L+1:t}\mathbf{W}_g\big), \quad \mathbf{S}_t = \text{Softmax}(\mathbf{H}_t\mathbf{W}_p), \tag{5}$$

where $\mathbf{W}_g$ and $\mathbf{W}_p$ are learnable weights, and $K \ll N$ denotes the number of communities, enforcing the low-rank constraint.

To guarantee that the partition effectively captures the manifold structure, we introduce two auxiliary losses adapted from differentiable graph pooling [3]. The *MinCut Loss* fosters strong intra-community connections within the stable topology, whereas the *Orthogonality Loss* ensures community distinctness (see Appendix B.1 for details):

$$\mathcal{L}_{\text{mincut}} = -\frac{\text{Tr}(\mathbf{S}_t^\top \mathbf{M}^t \mathbf{S}_t)}{\text{Tr}(\mathbf{S}_t^\top \mathbf{D}\mathbf{S}_t)}, \quad \mathcal{L}_{\text{orth}} = \left\| \frac{\mathbf{S}_t^\top \mathbf{S}_t}{\|\mathbf{S}_t^\top \mathbf{S}_t\|_F} - \frac{\mathbf{I}_K}{\sqrt{K}} \right\|^2. \tag{6}$$

3) Manifold Projection Implementation: We implement the operator defined in Proposition 4.1 via a pooling-broadcasting scheme. We first project $\mathbf{X}_{\text{diff},t}$ onto the manifold to obtain community representatives, and subsequently broadcast them back:

$$\mathbf{C}_{\text{va},t} = \text{Norm}_2\big(\mathbf{S}_t^\top \mathbf{X}_{\text{diff},t}\big), \quad \mathbf{X}_{\text{va},t} = \mathbf{S}_t \mathbf{C}_{\text{va},t}. \tag{7}$$

By strictly confining $\mathbf{X}_{\text{va},t}$ to the subspace spanned by $\mathbf{S}_t$, this process effectively filters out the high-rank idiosyncratic perturbations $\mathbf{N}_t$, isolating the purified systematic shifts $\mathbf{X}_{\text{va},t}$ for downstream.

4.2.2 ***Temporal Normalization.*** To mitigate the impact of intra-series shifts, we apply instance normalization to the extracted spatial invariant patterns $\mathbf{X}_{\text{inv},t}$. For each variate $i \in \{1, \dots, N\}$, we compute the temporal mean $\mu_{t,i}$ and standard deviation $\sigma_{t,i}$:

$$\mu_{t,i} = \frac{1}{L}\sum_{k=1}^{L} x_{\text{inv},t,i,k}, \quad \sigma_{t,i} = \sqrt{\frac{1}{L}\sum_{k=1}^{L}(x_{\text{inv},t,i,k} - \mu_{t,i})^2 + \epsilon}, \tag{8}$$

where ϵ is a small constant for numerical stability. Accordingly, the normalized vector $\hat{\mathbf{x}}_{\text{inv},t,i}$ is obtained by:

$$\hat{\mathbf{x}}_{\text{inv},t,i} = \frac{\mathbf{x}_{\text{inv},t,i} - \mu_{t,i}}{\sigma_{t,i}}. \tag{9}$$

4.2.3 ***Backbone Forecasting.*** The resulting $\hat{\mathbf{X}}_{\text{inv},t}$ serves as a spatiotemporally stationary representation, free from coupled distribution shifts. This representation is then fed into the backbone forecaster $\mathcal{F}_\psi$ to predict future states:

$$\hat{\mathbf{Y}}_{\text{inv},t} = \mathcal{F}_\psi(\hat{\mathbf{X}}_{\text{inv},t}). \tag{10}$$

Note that our framework is model-agnostic; $\mathcal{F}_\psi$ can be instantiated as any mainstream forecasting architecture.

4.3 Coupled Adaptive Denormalization

Standard denormalization simply reverses normalization statistics, which fails to capture dynamic future shifts and ignores the mutual influence between spatial and temporal dimensions. To address *Fallacy of Independent Restoration* identified in the Introduction, we propose the **Coupled Adaptive Denormalization (CAD)** module. Its primary objective is to achieve a holistic distribution restoration. Instead of treating inter- and intra-series shifts in isolation, we utilize the learned spatiotemporal coupling patterns as a bridge to jointly modulate the restoration process, generating adaptive parameters that recover the dynamic scale of the future horizon.

4.3.1 ***Spatio-Temporal Coupling Perception.*** As established in our analysis of the "Decoupled View", spatial structural evolutions and temporal dynamic drifts are inextricably coupled. To capture this, following [35], we propose a perception module designed to extract high-dimensional coupling patterns as discriminative cues for context-aware recalibration.

Patch-wise Temporal Segmentation. To discern the localized evolution of coupled shifts, we decompose the input series $\mathbf{X}_{t-L+1:t}$ into fine-grained temporal contexts. We partition $\mathbf{X}_{t-L+1:t}$ into $P = \lfloor L/L' \rfloor$ non-overlapping patches:

$$\mathbf{X}_{t-L+1:t} = \left[\mathbf{X}_t^{(1)}, \dots, \mathbf{X}_t^{(P)}\right], \quad \mathbf{X}_t^{(p)} \in \mathbb{R}^{N \times L'}, \tag{11}$$

where L' denotes the patch length. This discretization allows the model to capture how spatial correlations dynamically reconfigure within specific temporal windows, overcoming the limitations of static global dependency modeling.

Dynamic Coupling Graph Construction. Within each patch p, we instantiate the time-varying spatial coupling via a dynamic adjacency matrix. Let $\mathbf{x}_{t,i}^{(p)} \in \mathbb{R}^{L'}$ denote the temporal values of node i. We compute $\mathbf{A}_t^{(p)} \in \mathbb{R}^{N \times N}$ using the Pearson correlation coefficient to capture synchronized interaction patterns:

$$\mathbf{A}_{t,ij}^{(p)} = \text{Corr}\left(\mathbf{x}_{t,i}^{(p)}, \mathbf{x}_{t,j}^{(p)}\right). \tag{12}$$

Coupled Aggregation. To synthesize spatial structure with temporal dynamics, we employ a graph aggregation mechanism governed by the derived topology:

$$\hat{\mathbf{A}}_t^{(p)} = (\mathbf{D}_t^{(p)})^{-\frac{1}{2}}(\mathbf{A}_t^{(p)} + \mathbf{I})(\mathbf{D}_t^{(p)})^{-\frac{1}{2}}, \qquad \mathbf{H}_t^{(p)} = \hat{\mathbf{A}}_t^{(p)}\mathbf{X}_t^{(p)}. \tag{13}$$

Finally, we concatenate the localized representations to reconstruct the global coupling pattern $\mathbf{E}_{\text{ST},t}$:

$$\mathbf{E}_{\text{ST},t} = \text{Concat}\big(\mathbf{H}_t^{(1)}, \dots, \mathbf{H}_t^{(P)}\big) \in \mathbb{R}^{N \times L}. \tag{14}$$

4.3.2 ***Coupling-Aware Spatio-Temporal Denormalization.*** Simply addressing intra- and inter-series shifts sequentially is insufficient due to their complex entanglement. To overcome this, we design a mechanism that utilizes the learned coupling embedding $\mathbf{E}_{\text{ST},t}$ to dynamically regulate the restoration of both shift types.

Context-Aware Affine Transformation. To capture the evolving distribution shifts, we formulate the shift estimation as a conditional affine transformation process. This mechanism maps the global latent context to the required shift parameters. First, to obtain compact descriptors for this transformation, we distill the temporal statistics from both the spatio-temporal coupling pattern $\mathbf{E}_{\text{ST},t}$ and the purified variant features $\mathbf{X}_{\text{va},t}$ via temporal pooling:

$$\begin{aligned} \mathbf{e}_{\mu,t} &= \text{mean}_\tau(\mathbf{E}_{\text{ST},t}), \quad \mathbf{e}_{\sigma,t} = \text{std}_\tau(\mathbf{E}_{\text{ST},t}), \\ \mathbf{v}_{\mu,t} &= \text{mean}_\tau(\mathbf{X}_{\text{va},t}), \quad \mathbf{v}_{\sigma,t} = \text{std}_\tau(\mathbf{X}_{\text{va},t}). \end{aligned} \tag{15}$$

Based on these statistical descriptors, we define two parallel affine mappings to separately forecast the evolution of inter- and intra-series shifts. In our implementation, these affine transformations are empirically parameterized by MLPs, which project the concatenated condition vectors to the horizon-wise shift adjustments:

$$\begin{aligned} [\Delta\boldsymbol{\mu}_t^{\text{tem}},\ \mathbf{r}_t^{\text{tem}}] &= \gamma_{\text{tem}}([\boldsymbol{\mu}_t, \boldsymbol{\sigma}_t, \mathbf{e}_{\mu,t}, \mathbf{e}_{\sigma,t}]), \\ [\Delta\boldsymbol{\mu}_t^{\text{spa}},\ \mathbf{r}_t^{\text{spa}}] &= \gamma_{\text{spa}}([\mathbf{v}_{\mu,t}, \mathbf{v}_{\sigma,t}, \mathbf{e}_{\mu,t}, \mathbf{e}_{\sigma,t}]). \end{aligned} \tag{16}$$

The outputs represent the additive and multiplicative coefficients conditioned on both raw statistics and high-level coupling contexts.

Coupling-Gated Adaptive Fusion. A static combination of spatial and temporal branches is insufficient as the dominant factor varies dynamically (e.g., a traffic accident induces a spatial-dominant shift, while rush hour induces a temporal-dominant shift). Therefore, we employ a gating mechanism governed explicitly by the coupling statistics to adaptively balance the two branches:

$$\boldsymbol{\omega}_t = \text{Sigmoid}(\gamma_g([\mathbf{e}_{\mu,t}, \mathbf{e}_{\sigma,t}])), \tag{17}$$

where γ_g is implemented by MLPs. The fusion is performed as a convex combination, ensuring the final shift parameters reflect the dominant distribution driver:

$$\Delta\boldsymbol{\mu}_t = \boldsymbol{\omega}_t \odot \Delta\boldsymbol{\mu}_t^{\text{tem}} + (1 - \boldsymbol{\omega}_t) \odot \Delta\boldsymbol{\mu}_t^{\text{spa}}, \tag{18}$$

$$\mathbf{r}_t = \boldsymbol{\omega}_t \odot \mathbf{r}_t^{\text{tem}} + (1 - \boldsymbol{\omega}_t) \odot \mathbf{r}_t^{\text{spa}}. \tag{19}$$

Denormalization. To prevent gradient explosion and ensure training stability, we apply bounding functions $\mathcal{B}_\mu(\cdot)$ and $\mathcal{B}_\sigma(\cdot)$ to restrict the magnitude of the learned shift and scale adjustments (detailed in Appendix B.2). The final calibrated statistics $\tilde{\boldsymbol{\mu}}_t$ and $\tilde{\boldsymbol{\sigma}}_t$ are computed to restore the normalized prediction $\hat{\mathbf{Y}}_{\text{inv},t}$ to its original scale:

$$\tilde{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t + \mathcal{B}_\mu(\Delta\boldsymbol{\mu}_t), \quad \tilde{\boldsymbol{\sigma}}_t = \boldsymbol{\sigma}_t \odot \exp(\mathcal{B}_\sigma(\mathbf{r}_t)), \tag{20}$$

$$\hat{\mathbf{Y}}_{t+1:t+H} = \hat{\mathbf{Y}}_{\text{inv},t} \odot \tilde{\boldsymbol{\sigma}}_t + \tilde{\boldsymbol{\mu}}_t. \tag{21}$$

4.4 Optimization Objective

We train CAST-Norm end-to-end by minimizing the forecasting Mean Squared Error (MSE) over the prediction horizon H:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H} \left\| \hat{\mathbf{Y}}_{t+1:t+H} - \mathbf{X}_{t+1:t+H} \right\|^2. \tag{22}$$

To enforce the structural constraints grounded in the Manifold Hypothesis, we incorporate the regularization terms defined in the LMSD module. The total optimization objective is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_1 \big(\mathcal{L}_{\text{mincut}} + \mathcal{L}_{\text{orth}} \big) + \lambda_2 \mathcal{L}_{\text{stable}}, \tag{23}$$

where $\mathcal{L}_{\text{stable}}$ (Eq. 4) enforces topological consistency, and $\mathcal{L}_{\text{mincut}}$, $\mathcal{L}_{\text{orth}}$ (Eq. 6) optimize the quality of the low-rank community partition. Hyperparameters λ_1 and λ_2 balance forecasting accuracy with structural regularization.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate CAST-Norm on eight standard benchmarks: four **ETT** datasets (ETTh1/2, ETTm1/2) recording transformer temperatures [36], **Weather** [26] (meteorological indicators), **Exchange** [26] (daily currency rates), **Traffic** [26] (road occupancy), and **Electricity** [36] (hourly power consumption).

TSF Backbones. To validate the effectiveness of our framework, we integrate CAST-Norm with four classical backbones: DLinear [33] (Linear-based), Informer [36] (Transformer-based), Smamba [24] (Mamba-based) and TCN [1] (Convolution-based).

Normalization-based Baselines. To comprehensively validate the efficacy of our framework, we compare CAST-Norm against six state-of-the-art normalization-based baselines, including RevIN [13], DishTS [9], ST-Norm [6], FAN [30], SAN [18], and CAN-ST [29].

Evaluation Metrics. We evaluate forecasting performance using Mean Squared Error (MSE) and Mean Absolute Error (MAE), formulated as $\text{MSE} = \frac{1}{N}\sum_{t=1}^{N}(y_t - \hat{y}_t)^2$ and $\text{MAE} = \frac{1}{N}\sum_{t=1}^{N}|y_t - \hat{y}_t|$, where y_t and $\hat{y}_t$ denote the actual and forecasted values. MSE serves as our primary indicator. Due to space constraints, the supplementary tables for MAE results and standard deviations across three independent runs are provided in our code repository.

Implementation. We implemented our model and all baselines using PyTorch and conducted evaluations on a Linux server equipped with an NVIDIA A100 GPU. The experiments were performed under two distinct settings: *long-term* and *short-term* forecasting. For the long-term setting [36], we fixed the lookback window length at $L = 96$ and evaluated prediction horizons $H \in \{96, 192, 336, 720\}$. For the short-term setting [27, 31], both the lookback and prediction horizon were set to $L = H = 12$. Following standard protocols established in previous works [4, 37], the ETT datasets were partitioned into training, validation, and test sets with a 6:2:2 ratio. For other datasets, we adopted the standard 7:1:2 split.

5.2 Overall Performance

Table 1 summarizes the comparison between CAST-Norm and state-of-the-art normalization-based frameworks across eight datasets and four diverse forecasting backbones. The results demonstrate that our method consistently surpasses the baselines in the majority of scenarios, validating its robustness across varying model architectures and data scales. This performance improvement is primarily attributed to two factors: (1) In the normalization phase, we simultaneously account for both inter- and intra-series shifts, thereby providing the forecasting backbone with stationary spatiotemporal patterns. (2) In the denormalization phase, by explicitly incorporating the impact of spatiotemporal coupling, we achieve a more precise adaptive distribution restoration.

5.3 Efficiency Analysis

We analyze the computational complexity of CAST-Norm, where N, L, H, and K denote the number of variates, lookback window size, prediction horizon, and communities, respectively.

Theoretical Complexity. The time overhead is dominated by pairwise correlation computation and graph-based projection in the normalization phase, scaling as $O(N^2L)$, while denormalization scales linearly as $O(N(L+H))$. The space overhead requires $O(N^2)$ for topology memory and $O(NL)$ for intermediate representations, totaling $O(N^2 + NL)$.

Empirical Efficiency. We evaluated computational costs on the Traffic dataset using S-Mamba across short- and long-term settings (Table 2). Although the $O(N^2L)$ complexity is theoretically non-trivial, practical overhead remains highly acceptable. Specifically, CAST-Norm utilizes only 7.63–7.97M parameters and requires 40.60–60.02 seconds per epoch. This confirms our module maintains a footprint comparable to baselines without introducing significant latency. For deployments involving extremely large-scale variables, efficiency can be further optimized by integrating approximation strategies such as sparse attention [36] or subgraph sampling [34].

5.4 Ablation Studies

To rigorously evaluate the contribution of each component within the CAST-Norm framework, we conducted a comprehensive ablation study. Due to space constraints, we present the results for the ETTm2 dataset with the DLinear backbone in Figure 2.

1) Necessity of Inter-Series Shift Mitigation (CAST-I). This variant removes the components related to inter-series shift, degrading the framework to standard RevIN. The performance drop confirms that focusing exclusively on intra-series shift is insufficient, and explicitly modeling inter-series shift is critical for MTSF.

2) Effectiveness of Low-rank Variant Pattern Purification (CAST-L). This variant replaces the purified variant patterns $\mathbf{X}_{\text{va},t}$ with raw residuals $\mathbf{X}_{\text{diff},t}$, bypassing the low-rank manifold projection. The inferior results validate that raw residuals contain unlearnable stochastic noise that biases the estimator, necessitating our purification strategy to isolate systematic spatial evolutions.

3) Impact of Spatio-Temporal Coupling Perception (CAST-C, CAST-R, & CAST-R-L). To thoroughly validate the necessity and

Table 1: Overall performance (MSE) comparison. '/' denotes forecasting performance from the backbone alone. For each setting, the best result is highlighted in bold with a violet background. All values are averaged over three runs.

	Models	ETTm1					ETTm2					ETTh1					ETTh2				
		Short	Long				Short	Long				Short	Long				Short	Long			
		12	96	192	336	720	12	96	192	336	720	12	96	192	336	720	12	96	192	336	720
DLinear	/	0.322	0.356	0.392	0.424	0.483	0.096	0.193	0.281	0.382	0.533	0.810	0.399	0.455	0.512	0.514	0.212	0.345	0.478	0.592	0.828
	RevIN	0.343	0.354	0.391	0.424	0.486	0.097	0.188	0.255	0.337	0.428	0.819	0.399	**0.449**	0.518	0.529	0.202	0.307	0.396	0.441	0.618
	DishTS	0.313	0.362	0.399	0.425	0.485	0.099	0.208	0.306	0.416	0.605	0.775	0.408	0.458	0.512	0.545	0.198	0.371	0.512	0.591	0.828
	ST-Norm	0.297	0.365	0.394	0.423	0.498	0.099	0.195	0.283	0.389	0.535	0.772	0.404	0.454	0.502	0.544	0.194	0.338	0.466	0.573	0.788
	FAN	0.302	0.362	0.400	0.435	0.499	0.096	0.197	0.281	0.382	0.538	**0.651**	0.423	0.480	0.527	0.582	0.196	0.366	0.481	0.579	0.793
	SAN	0.298	0.354	0.390	0.422	0.483	0.096	0.193	0.281	0.380	0.537	0.681	0.403	0.452	0.501	0.538	0.194	0.338	0.469	0.574	0.790
	CAN-ST	0.301	0.358	0.393	0.427	0.491	0.096	0.215	0.321	0.438	0.754	0.724	0.406	0.455	0.505	0.546	0.198	0.337	0.513	0.599	0.887
	CAST-Norm	**0.294**	**0.351**	**0.387**	**0.420**	**0.481**	**0.094**	**0.184**	**0.247**	**0.307**	**0.408**	0.721	**0.398**	0.451	**0.500**	**0.509**	**0.191**	**0.305**	**0.388**	**0.424**	**0.606**
Informer	/	0.227	0.665	0.674	1.086	1.085	0.145	0.418	0.636	1.446	3.707	0.612	0.913	0.977	1.082	**1.208**	0.250	2.985	6.061	4.864	4.119
	RevIN	0.227	0.484	0.583	0.716	0.842	0.098	0.453	0.559	0.761	2.018	0.517	0.744	0.895	1.054	1.266	0.180	**0.426**	1.031	1.294	1.701
	DishTS	0.242	0.504	0.610	0.769	0.896	0.136	0.568	0.754	1.124	3.696	0.531	0.969	0.999	1.167	1.284	0.367	2.046	5.091	3.806	4.362
	ST-Norm	0.220	0.474	0.579	0.720	0.825	0.100	0.398	0.532	0.765	2.012	0.519	0.745	0.898	1.052	1.275	0.180	0.472	0.975	1.248	1.635
	FAN	0.232	0.476	0.581	0.725	0.828	0.099	0.401	0.581	0.983	2.397	0.630	0.751	0.902	1.061	1.272	0.224	1.488	2.858	2.732	2.749
	SAN	**0.212**	0.475	0.578	0.722	0.824	0.098	0.487	0.691	0.922	2.008	0.515	0.748	0.899	1.055	1.269	0.181	0.623	1.614	1.571	1.638
	CAN-ST	0.240	0.478	0.592	0.728	0.827	0.099	0.466	0.728	1.131	2.015	0.543	0.755	0.959	1.057	1.271	0.232	1.172	1.959	1.969	2.779
	CAST-Norm	0.215	**0.472**	**0.571**	**0.715**	**0.818**	**0.097**	**0.390**	**0.526**	**0.758**	**1.994**	**0.511**	**0.738**	**0.893**	**1.046**	1.266	**0.179**	0.469	**0.971**	**1.242**	**1.631**
S-Mamba	/	0.249	0.412	0.451	0.587	0.665	0.110	0.479	0.845	1.814	3.501	0.564	0.467	0.462	0.534	0.593	0.237	1.513	2.148	2.397	2.211
	RevIN	0.242	0.349	0.415	0.456	0.596	0.086	0.229	0.436	0.703	0.956	0.516	0.395	0.472	0.525	0.603	0.166	0.295	**0.458**	0.533	0.762
	DishTS	0.231	0.397	0.415	0.449	0.565	0.087	0.263	0.538	0.702	1.017	0.541	0.398	0.472	0.535	0.596	0.170	0.370	0.569	1.048	0.957
	ST-Norm	0.212	0.348	0.411	0.458	0.562	0.089	0.226	0.421	0.699	0.958	0.506	0.395	0.469	0.531	0.602	0.167	0.354	0.462	0.535	0.761
	FAN	0.259	0.385	0.443	0.488	0.568	**0.083**	0.225	0.419	0.705	0.962	0.576	0.447	0.501	0.581	0.657	**0.165**	0.362	0.504	0.622	0.855
	SAN	0.225	0.349	0.412	0.444	0.561	0.088	0.224	0.418	0.697	0.955	0.505	0.394	0.499	0.561	0.631	0.170	0.351	0.460	0.533	0.760
	CAN-ST	0.219	0.351	0.415	0.445	0.564	0.086	0.226	0.420	0.701	0.960	0.506	0.396	0.471	0.534	0.598	0.173	0.379	0.538	0.640	0.758
	CAST-Norm	**0.200**	**0.344**	**0.406**	**0.442**	**0.559**	0.085	**0.222**	**0.417**	**0.694**	**0.952**	**0.504**	**0.393**	**0.467**	**0.528**	**0.590**	**0.165**	**0.349**	0.459	**0.532**	**0.756**
TCN	/	0.263	0.570	0.598	0.676	0.777	0.129	0.519	0.726	1.334	3.418	0.645	0.713	0.825	0.901	0.995	0.258	1.769	3.581	3.918	3.821
	RevIN	0.255	0.353	0.413	0.460	0.493	0.085	0.266	0.404	0.646	2.027	0.516	0.421	0.464	0.702	0.843	0.158	0.377	0.669	0.828	1.441
	DishTS	0.234	0.394	0.470	0.530	0.655	0.142	0.385	0.451	0.869	1.997	0.569	0.524	0.578	0.702	0.845	0.342	0.764	1.730	2.179	2.233
	ST-Norm	0.237	0.352	**0.399**	0.454	0.585	0.088	0.264	0.403	0.621	1.996	0.512	0.446	0.569	0.701	0.843	0.158	0.369	0.655	0.812	1.290
	FAN	0.240	0.412	0.451	0.509	0.583	0.085	0.269	0.411	0.652	1.995	0.530	0.434	0.501	0.702	0.855	0.165	0.369	0.677	0.817	1.393
	SAN	0.226	0.351	0.401	0.454	0.540	0.089	0.263	0.408	0.628	2.010	0.497	**0.411**	**0.456**	0.705	0.872	0.165	0.485	0.692	0.811	1.288
	CAN-ST	0.244	0.349	0.412	0.454	0.547	0.085	0.264	0.403	0.621	1.996	0.499	0.446	0.569	0.703	0.842	0.164	0.369	0.653	0.811	1.336
	CAST-Norm	**0.222**	**0.349**	**0.399**	**0.452**	**0.538**	**0.084**	**0.262**	**0.401**	**0.619**	**1.994**	**0.486**	0.444	0.567	**0.699**	**0.841**	**0.157**	**0.367**	**0.651**	**0.809**	**1.285**

	Models	Weather					Traffic					Exchange					Electricity				
		Short	Long				Short	Long				Short	Long				Short	Long			
		12	96	192	336	720	12	96	192	336	720	12	96	192	336	720	12	96	192	336	720
DLinear	/	0.090	0.196	0.237	0.282	0.346	1.240	0.696	0.646	0.653	0.694	0.039	0.096	0.186	0.323	**0.765**	0.662	0.210	0.210	0.223	0.258
	RevIN	0.088	0.196	0.240	0.292	0.365	1.158	0.691	0.644	0.651	0.689	0.018	0.089	0.180	0.324	0.854	0.774	0.213	0.213	0.227	0.268
	DishTS	0.093	0.199	0.237	0.286	0.355	1.160	0.695	0.652	0.658	0.696	0.018	0.116	0.249	0.489	1.097	0.590	0.204	0.208	0.221	0.256
	ST-Norm	0.092	0.198	0.236	0.282	0.346	1.165	0.683	0.637	0.644	0.681	0.018	0.085	0.169	0.301	0.848	0.654	0.208	0.208	0.222	0.260
	FAN	0.094	0.197	0.238	0.284	0.344	**0.910**	0.675	0.637	0.658	0.715	0.024	0.101	0.210	0.417	0.949	0.588	0.203	0.207	0.223	0.260
	SAN	0.089	0.196	0.239	0.283	0.347	1.102	0.676	0.630	0.632	0.673	0.018	0.086	0.182	0.328	0.930	0.586	0.203	0.204	0.216	0.251
	CAN-ST	0.093	0.196	0.236	0.282	0.356	1.125	0.676	0.629	0.635	0.675	0.021	0.121	0.296	0.564	1.116	0.590	0.204	0.203	0.217	0.252
	CAST-Norm	**0.087**	**0.195**	**0.235**	**0.281**	**0.345**	1.167	**0.672**	**0.627**	**0.631**	**0.672**	**0.017**	**0.081**	**0.165**	**0.293**	0.777	**0.570**	**0.202**	**0.202**	**0.215**	**0.250**
Informer	/	0.097	0.372	0.458	0.640	0.988	0.597	0.718	0.752	0.840	0.995	0.306	0.946	1.087	1.599	2.752	0.245	0.340	0.357	0.359	0.394
	RevIN	0.091	0.226	0.324	0.388	0.542	1.158	0.695	0.752	0.866	1.031	0.016	0.183	0.441	0.685	**1.811**	0.237	0.213	0.232	0.278	0.323
	DishTS	0.095	0.383	0.519	0.614	0.933	0.991	0.704	0.740	0.862	1.026	0.023	0.233	0.466	1.069	1.985	0.246	0.244	0.257	0.285	0.346
	ST-Norm	0.094	0.228	0.325	0.389	0.543	1.134	0.691	0.764	0.867	1.039	0.016	0.187	0.452	0.689	2.050	0.225	0.214	0.230	0.277	0.322
	FAN	0.099	0.279	0.399	0.519	0.665	0.825	0.680	0.735	0.850	0.944	0.084	0.733	1.261	1.479	2.242	0.191	0.215	0.231	0.282	0.325
	SAN	0.091	0.230	0.350	0.454	0.548	0.830	0.685	0.738	0.852	0.948	0.016	0.208	0.448	0.798	2.105	0.188	0.250	0.231	0.279	0.328
	CAN-ST	0.091	0.300	0.407	0.533	0.649	0.825	0.680	0.736	0.849	0.945	0.031	0.396	0.730	1.072	2.246	0.195	0.214	0.236	0.279	0.329
	CAST-Norm	**0.090**	**0.225**	**0.323**	**0.387**	**0.541**	**0.822**	**0.679**	**0.734**	**0.849**	**0.943**	**0.015**	**0.182**	**0.439**	**0.683**	2.225	**0.187**	**0.212**	**0.229**	**0.276**	**0.321**
S-Mamba	/	**0.075**	0.172	0.219	0.281	0.382	0.650	0.560	0.578	0.601	0.640	0.047	0.451	1.236	1.796	2.563	**0.209**	0.154	0.169	0.192	0.232
	RevIN	0.080	0.167	0.216	0.280	0.357	0.791	0.440	0.459	0.481	0.525	0.015	0.101	0.387	0.582	0.922	0.215	0.151	0.168	0.185	0.226
	DishTS	0.080	0.168	0.216	0.280	0.354	0.717	0.445	0.462	0.485	0.528	0.015	0.105	0.392	0.588	1.110	0.238	0.150	0.165	0.183	0.212
	ST-Norm	0.082	0.166	0.211	0.280	0.337	0.713	0.442	0.460	0.488	0.526	0.015	0.101	0.388	0.582	0.923	0.236	0.149	0.165	0.181	0.212
	FAN	0.090	0.222	0.273	0.358	0.537	0.849	0.561	0.568	0.598	0.678	0.019	0.130	0.405	0.589	1.097	0.323	0.194	0.202	0.221	0.258
	SAN	0.082	0.166	0.218	0.282	0.352	0.685	0.440	0.459	0.487	0.529	0.015	0.104	0.391	0.584	1.081	0.218	0.157	0.181	0.198	0.223
	CAN-ST	0.083	0.166	0.211	0.279	0.423	0.680	0.455	0.479	0.504	0.557	0.021	0.141	0.395	0.582	1.272	0.223	0.149	0.165	0.182	**0.210**
	CAST-Norm	0.081	**0.165**	**0.210**	**0.279**	**0.336**	**0.679**	**0.439**	**0.458**	**0.480**	**0.524**	**0.014**	**0.100**	**0.386**	**0.581**	**0.921**	0.221	**0.148**	**0.164**	**0.180**	0.211
TCN	/	0.091	0.223	0.298	0.406	0.787	0.646	0.759	0.745	0.759	0.789	0.142	0.678	0.916	1.150	1.998	0.275	0.313	0.323	0.328	0.331
	RevIN	0.075	0.168	0.217	0.277	0.363	0.850	0.691	0.713	0.736	0.782	**0.013**	0.117	0.197	0.500	1.026	0.200	0.203	0.217	0.248	0.266
	DishTS	0.078	0.180	0.259	0.308	0.384	0.984	0.696	0.712	0.732	0.782	0.018	0.172	0.273	0.501	0.783	0.274	0.238	0.249	0.256	0.273
	ST-Norm	0.075	0.168	0.214	0.270	0.363	1.222	0.689	0.710	0.733	0.782	0.014	0.117	0.197	0.500	0.783	0.252	0.201	0.214	0.248	0.266
	FAN	0.138	0.302	0.463	0.567	0.606	0.711	0.645	0.671	0.696	0.783	0.019	0.136	0.303	0.583	1.013	0.235	0.200	0.213	0.248	0.266
	SAN	0.076	0.167	0.215	0.275	0.384	0.815	0.645	0.671	0.696	0.782	0.014	0.117	0.231	0.501	0.975	**0.188**	0.200	0.215	0.249	0.267
	CAN-ST	0.076	0.187	0.274	0.449	0.762	**0.625**	0.645	0.671	0.696	0.782	0.020	0.156	0.358	0.663	1.228	0.200	0.200	0.213	0.248	0.266
	CAST-Norm	**0.074**	**0.167**	**0.213**	**0.266**	**0.362**	0.903	**0.644**	**0.670**	**0.695**	**0.781**	0.014	**0.116**	**0.196**	**0.499**	**0.782**	0.209	**0.199**	**0.212**	**0.247**	**0.265**

effectiveness of our coupling-aware joint denormalization, we design three variants. First, **CAST-C** removes the coupling perception and performs temporal and spatial denormalization sequentially and independently. Its suboptimal performance reveals that sequential operations apply the coupled shift twice. This redundant "double-shift" bias compounds over time, explaining why the absence of joint restoration disproportionately degrades long-horizon predictions. Second, **CAST-R** performs joint spatiotemporal denormalization but removes the explicit coupling guidance. Third, **CAST-R-L** builds upon CAST-R by vastly enlarging its hidden dimensions, ensuring its total parameter count strictly exceeds

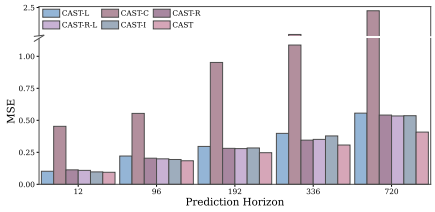


Figure 2: Performance comparison between CAST-Norm and its variants on the ETTm2 dataset with DLinear.

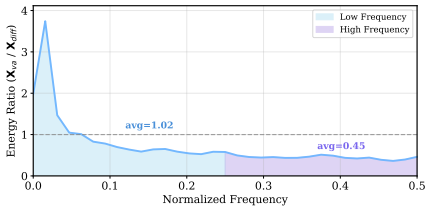


Figure 3: Spectral energy ratio comparison between purified variant and residual on the Weather dataset with DLinear.

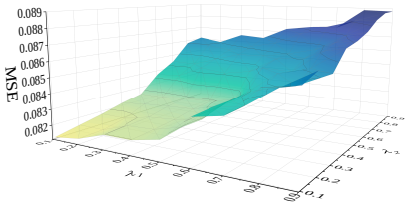


Figure 4: Performance *w.r.t.* regularization weights λ_1 and λ_2 evaluated on the Exchange dataset.

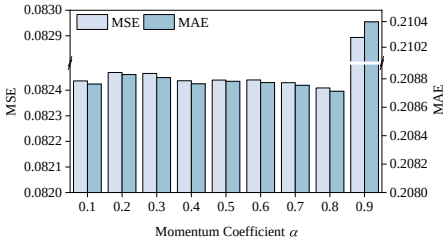


Figure 5: Performance *w.r.t.* EMA momentum α with DLinear as backbone evaluated on the Exchange dataset.

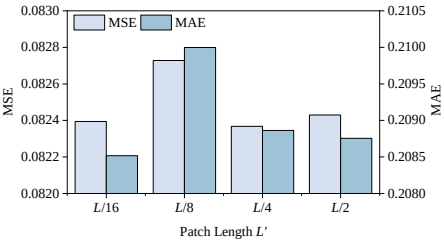


Figure 6: Performance *w.r.t.* patch length L' with DLinear as backbone evaluated on the Exchange dataset.

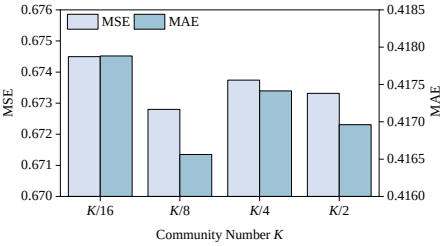


Figure 7: Performance *w.r.t.* community number K with DLinear as backbone evaluated on the Traffic dataset.

Table 2: Comparisons of training/inference time costs and parameter volumes on the Traffic dataset using the S-Mamba backbone. T Speed and I Speed represent training speed (seconds per epoch) and inference speed (seconds per run). Short-term results are evaluated at prediction length 12, while long-term results are averaged over lengths {96, 192, 336, 720}.

Model	T Speed (s/ep)	Parameters	I Speed (s/run)
Short-term Forecasting (H = 12)			
RevIN	32.1592	7,607,628	4.4231
DishTS	32.5564	7,630,040	4.5475
ST-Norm	32.6810	7,611,076	4.7621
FAN	42.3911	7,621,588	6.5894
SAN	33.0351	7,626,762	4.3891
CAN-ST	43.6037	7,739,082	6.3855
CAST-Norm	40.5952	7,632,097	5.4716
Long-term Forecasting (Avg. over 96, 192, 336, 720)			
RevIN	36.4103	7,816,848	10.0289
DishTS	35.8895	7,984,076	9.8283
ST-Norm	54.7490	7,820,296	15.5349
FAN	38.4409	7,888,732	11.1534
SAN	37.3338	7,984,516	10.4051
CAN-ST	41.3755	7,948,302	10.7785
CAST-Norm	60.0168	7,970,305	17.7349

that of CAST-Norm. Results show that CAST-R noticeably lags behind CAST-Norm due to the absence of precise coupling guidance. Furthermore, CAST-R-L yields no additional performance benefit despite the significant increase in parameters. Together, these three variants strongly validate that our improvements stem explicitly from effective spatiotemporal coupling modeling rather than mere increases in architectural capacity.

5.5 Effectiveness Analysis of Variant Pattern Purification

To validate that the low-rank manifold projection effectively suppresses high-frequency noise while preserving low-frequency signals, we conduct a spectral analysis using 200 randomly sampled test windows (length 96) from the Weather dataset with the DLinear backbone. We estimate the Power Spectral Density (PSD) using Welch's method [25] (segment length 32, 50% overlap) to compute the frequency-wise energy ratio $R(f) = \frac{\mathrm{PSD}(\mathbf{X}_{\mathrm{va}})}{\mathrm{PSD}(\mathbf{X}_{\mathrm{diff}})}$ over the normalized spectrum $[0, 0.5]$ (up to the Nyquist frequency [21]). Following standard frequency domain analysis [11], we bisect the spectrum into a low-frequency band $[0, 0.25]$ (systematic patterns) and a high-frequency band $(0.25, 0.5]$ (stochastic perturbations). As shown in Figure 3, the energy ratio remains close to 1.0 in the low-frequency region, indicating that systematic evolutionary patterns are accurately preserved. Conversely, the ratio drops significantly in the high-frequency region, demonstrating effective suppression of stochastic perturbations. These results confirm that LMSD successfully filters random noise while retaining meaningful shift information.

5.6 Hyperparameter Analysis

We investigate the sensitivity of CAST-Norm to critical hyperparameters. This analysis identifies the optimal operating ranges for

each component and verifies the model's robustness against parameter variations. To ensure a comprehensive evaluation, we utilize the **Exchange** dataset for the experiments regarding regularization weights (§5.6.1), EMA momentum (§5.6.2), and patch length (§5.6.3). For the sensitivity analysis of the community number (§5.6.4), we employ the **Traffic** dataset. Since Traffic possesses the highest dimensionality (largest node count), it provides the most challenging scenario to rigorously evaluate the impact of community partitioning. All experiments are conducted with a fixed input length of 96 and a prediction horizon of 96.

5.6.1 ***Sensitivity of Regularization Weights.*** We analyze λ_1 and λ_2 by varying them across $\{0.1, 0.2, \ldots, 0.9\}$. Excessively large weights tend to dominate the total loss, diverting optimization focus from the primary forecasting objective. Conversely, overly small weights fail to enforce necessary structural constraints, leading to noisy or insufficient spatial disentanglement. Figure 4 confirms the model's robustness, demonstrating consistently superior performance when both λ_1 and λ_2 are within the range of $[0.1, 0.5]$.

5.6.2 ***Sensitivity of EMA Momentum.*** We analyze the momentum α within the range $[0.1, 0.9]$, which balances memory stability and adaptivity. Small α introduces transient noise, while large α induces inertia and lagging. As shown in Figure 5, the model is robust within $\alpha \in [0.4, 0.8]$, confirming that this range optimally captures long-term stable dependencies without succumbing to instantaneous fluctuations.

5.6.3 ***Sensitivity of Patch Length.*** We investigate the patch length L' over the set $\{L/2, L/4, L/8, L/16\}$, which dictates the granularity of the coupling graph. While excessively long patches oversmooth dynamics and short patches lack sufficient samples, Figure 6 indicates that the model is generally robust to L' variations. Specifically, setting $L' \in \{L/16, L/4\}$ achieves the best trade-off between dynamic sensitivity and estimation reliability.

5.6.4 ***Sensitivity of Community Number.*** We examine the community number K across $\{N/2, N/4, N/8, N/16\}$, which determines the manifold rank. An overly small K causes underfitting by merging distinct factors, whereas an excessive K fails to filter high-frequency stochastic perturbations. Experimental results in Figure 7 demonstrate that performance remains stable across settings, with $K \approx N/8$ optimally defining latent factors to isolate systematic shifts.

5.7 Robustness against Distribution Shifts

To systematically simulate varying degrees of distribution shift, following [15], we use a noise interpolation mechanism based on additive Gaussian noise. Given the original time series input $\mathbf{X} \in \mathbb{R}^{N \times T}$, where N and T denote the number of variates and time steps respectively, we generate a noise matrix $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ of the same shape as $\mathbf{X}$. To simulate different shift intensities, we linearly interpolate between the original data and the noise using a mixing coefficient $a \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$. The noisy input is then computed as $\mathbf{X}_{\text{noisy}} = (1 - a) \cdot \mathbf{X} + a \cdot \boldsymbol{\epsilon}$, where a higher a indicates a stronger distribution shift. We evaluated the robustness of our proposed method against other representative normalization techniques (SAN and FAN) using the DLinear backbone on the ETTh2 dataset with a prediction length of $H = 96$. The synthetic out-of-distribution (OOD) noise was injected during the inference phase to strictly test the models' generalization capabilities under sudden regime shifts.

As demonstrated in Table 3, while the performance of all models naturally degrades as the shift intensity (a) increases, CAST-Norm consistently maintains the lowest forecasting error across all severity levels. This controlled evaluation rigorously confirms that explicit modeling of spatiotemporal coupling provides superior robustness and stability when confronting severe, unexpected distribution shifts in real-world scenarios.

Table 3: Forecasting performance (MSE) under varying synthetic distribution shift intensities on ETTh2 dataset ($H = 96$).

Model / Shift Intensity (a)	0.0	0.2	0.4	0.6	0.8
DLinear	0.345	0.591	0.983	1.546	2.278
+ SAN	0.338	0.438	0.804	1.396	2.213
+ FAN	0.366	0.651	1.048	1.615	2.360
+ CAST-Norm (Ours)	**0.305**	**0.397**	**0.683**	**1.024**	**1.938**

6 Conclusion

This paper challenges the prevalent **"Decoupled View"** in MTSF, identifying it as a theoretical bottleneck that erroneously treats intra- and inter-series shifts as orthogonal components. We theoretically establish that these shifts are **mutually coupled**, necessitating a paradigm transition toward holistic modeling. To realize this, we propose **Coupled Adaptive Spatio-Temporal Normalization** (CAST-Norm). Our *Hybrid Inter-Intra Series Normalization* employs a critical ordering principle to simultaneously yield rigorously stationary patterns for forecasting and isolate systematic evolutions. Specifically, the incorporated *Low-rank Manifold Spatial Decoupling* module exploits the Manifold Hypothesis to filter stochastic perturbations, ensuring precise extraction of these systematic shifts. Furthermore, to resolve the "Fallacy of Independent Restoration," our *Coupled Adaptive Denormalization* leverages spatiotemporal entanglement to *jointly modulate* future distribution recovery. Extensive experiments across eight datasets and four backbones demonstrate that CAST-Norm consistently achieves state-of-the-art performance. Moving forward, a promising direction is to extend CAST-Norm to explicit Spatio-Temporal Forecasting (STF) by developing graph-guided mechanisms that elegantly fuse our dynamic coupled normalization with predefined physical spatial priors. Such an extension would unlock broader applicability to high-impact STF scenarios such as traffic forecasting and urban sensing, where reconciling data-driven manifold coupling with domain-specific topological knowledge holds strong potential for advancing predictive accuracy.

Acknowledgments

This research is funded by the Science and Technology Development Fund (FDCT), Macau SAR (file no. 0123/2023/RIA2, 001/2024/SKL 0002/2025/EQP, 0072/2025/AMJ), and the Research Grant of University of Macau (File no. MYRG-GRG2025-00241-IOTSC, APAEM/SG/00037/2026).

References

[1] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. 2018. An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. arXiv:1803.01271 [cs.LG] https://arxiv.org/abs/1803.01271

[2] Mikhail Belkin and Partha Niyogi. 2003. Laplacian Eigenmaps for dimensionality reduction and data representation. *Neural Comput.* 15, 6 (June 2003), 1373–1396. doi:10.1162/089976603321780317

[3] Filippo Maria Bianchi, Daniele Grattarola, and Cesare Alippi. 2020. Spectral clustering with graph neural networks for graph pooling. In *Proceedings of the 37th International Conference on Machine Learning (ICML'20)*. JMLR.org, Article 82, 10 pages.

[4] Wanlin Cai, Yuxuan Liang, Xianggen Liu, Jianshuai Feng, and Yuankai Wu. 2024. MSGNet: learning multi-scale inter-series correlations for multivariate time series forecasting. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence (AAAI'24/IAAI'24/EAAI'24)*. AAAI Press, Article 1243, 9 pages. doi:10.1609/aaai.v38i10.28991

[5] Emmanuel J. Candès, Xiaodong Li, Yi Ma, and John Wright. 2011. Robust principal component analysis? *J. ACM* 58, 3, Article 11 (June 2011), 37 pages. doi:10.1145/1970392.1970395

[6] Jinliang Deng, Xiusi Chen, Renhe Jiang, Xuan Song, and Ivor W. Tsang. 2021. ST-Norm: Spatial and Temporal Normalization for Multi-variate Time Series Forecasting. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining* (Virtual Event, Singapore) *(KDD '21)*. Association for Computing Machinery, New York, NY, USA, 269–278. doi:10.1145/3447548.3467330

[7] Yuntao Du, Jindong Wang, Wenjie Feng, Sinno Pan, Tao Qin, Renjun Xu, and Chongjun Wang. 2021. AdaRNN: Adaptive Learning and Forecasting of Time Series. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management* (Virtual Event, Queensland, Australia) *(CIKM '21)*. Association for Computing Machinery, New York, NY, USA, 402–411. doi:10.1145/3459637.3482315

[8] Wenying Duan, Xiaoxi He, Lu Zhou, Lothar Thiele, and Hong Rao. 2023. Combating distribution shift for accurate time series forecasting via hypernetworks. In *2022 IEEE 28th International Conference on Parallel and Distributed Systems (ICPADS)*. IEEE, 900–907.

[9] Wei Fan, Pengyang Wang, Dongkun Wang, Dongjie Wang, Yuanchun Zhou, and Yanjie Fu. 2023. Dish-TS: A General Paradigm for Alleviating Distribution Shift in Time Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 7522–7529.

[10] Yuchen Fang, Yuxuan Liang, Bo Hui, Zezhi Shao, Liwei Deng, Xu Liu, Xinke Jiang, and Kai Zheng. 2025. Efficient Large-Scale Traffic Forecasting with Transformers: A Spatial Data Management Perspective. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1* (Toronto ON, Canada) *(KDD '25)*. Association for Computing Machinery, New York, NY, USA, 307–317. doi:10.1145/3690624.3709177

[11] Jingru Fei, Kun Yi, Wei Fan, Qi Zhang, and Zhendong Niu. 2025. Amplifier: bringing attention to neglected low-energy components in time series forecasting. In *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence and Thirty-Seventh Conference on Innovative Applications of Artificial Intelligence and Fifteenth Symposium on Educational Advances in Artificial Intelligence (AAAI'25/IAAI'25/EAAI'25)*. AAAI Press, Article 1294, 9 pages. doi:10.1609/aaai.v39i11.33267

[12] Hui He, Qi Zhang, Kun Yi, Xiaojun Xue, Shoujin Wang, Liang Hu, and Longbing Cao. 2024. Robust Multivariate Time Series Forecasting against Intra-and Inter-Series Transitional Shift. *arXiv preprint arXiv:2407.13194* (2024).

[13] Taesung Kim, Jinhee Kim, Yunwon Tae, Cheonbok Park, Jang-Ho Choi, and Jaegul Choo. 2021. Reversible Instance Normalization for Accurate Time-Series Forecasting against Distribution Shift. In *International Conference on Learning Representations*. https://openreview.net/forum?id=cGDAkQo1C0p

[14] Yuxuan Liang, Yutong Xia, Songyu Ke, Yiwei Wang, Qingsong Wen, Junbo Zhang, Yu Zheng, and Roger Zimmermann. 2022. AirFormer: Predicting Nationwide Air Quality in China with Transformers. *arXiv preprint arXiv:2211.15979* (2022).

[15] Chi-Heng Lin, Chiraag Kaushik, Eva L. Dyer, and Vidya Muthukumar. 2024. The good, the bad and the ugly sides of data augmentation: An implicit spectral regularization perspective. *Journal of Machine Learning Research* 25, 91 (2024), 1–85. http://jmlr.org/papers/v25/22-1312.html

[16] Yong Liu, Chenyu Li, Jianmin Wang, and Mingsheng Long. 2023. Koopa: Learning Non-stationary Time Series Dynamics with Koopman Predictors. In *Thirty-seventh Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=A4zzxu82a7

[17] Yong Liu, Haixu Wu, Jianmin Wang, and Mingsheng Long. 2022. Non-stationary Transformers: Exploring the Stationarity in Time Series Forecasting. (2022).

[18] Zhiding Liu, Mingyue Cheng, Zhi Li, Zhenya Huang, Qi Liu, Yanhu Xie, and Enhong Chen. 2023. Adaptive normalization for non-stationary time series forecasting: a temporal slice perspective. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 628, 20 pages.

[19] Boris N. Oreshkin, Dmitri Carpov, Nicolas Chapados, and Yoshua Bengio. 2020. N-BEATS: Neural basis expansion analysis for interpretable time series forecasting. In *International Conference on Learning Representations*. https://openreview.net/forum?id=r1ecqn4YwB

[20] Chao Shang, Jie Chen, and Jinbo Bi. 2021. Discrete Graph Structure Learning for Forecasting Multiple Time Series. In *International Conference on Learning Representations*. https://openreview.net/forum?id=WEHSlH5mOk

[21] C.E. Shannon. 1949. Communication in the Presence of Noise. *Proceedings of the IRE* 37, 1 (1949), 10–21. doi:10.1109/JRPROC.1949.232969

[22] Artyom Stitsyuk and Jaesik Choi. 2025. xPatch: Dual-Stream Time Series Forecasting with Exponential Seasonal-Trend Decomposition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 20601–20609.

[23] Yuyang Wang, Alex Smola, Danielle C Maddix, Jan Gasthaus, Bernie Wang, and Ali Caner Turkmen. 2019. Deep factors for forecasting. In *International Conference on Machine Learning*. PMLR, 6607–6617.

[24] Zihan Wang, Fanheng Kong, Shi Feng, Ming Wang, Xiaocui Yang, Han Zhao, Daling Wang, and Yifei Zhang. 2025. Is Mamba effective for time series forecasting? *Neurocomput.* 619, C (Feb. 2025), 14 pages. doi:10.1016/j.neucom.2024.129178

[25] P. Welch. 1967. The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics* 15, 2 (1967), 70–73. doi:10.1109/TAU.1967.1161901

[26] Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. 2021. Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting. In *Advances in Neural Information Processing Systems*.

[27] Haihua Xu, Wei Fan, Kun Yi, and Pengyang Wang. 2024. Decoupled Invariant Attention Network for Multivariate Time-series Forecasting. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, Kate Larson (Ed.). International Joint Conferences on Artificial Intelligence Organization, 2487–2495. doi:10.24963/ijcai.2024/275 Main Track.

[28] Haihua Xu, Qi Hao, He Zhang, Jianpeng Zhao, Ziyue Qiao, Lu Jiang, Pengfei Wang, Yingjie Zhou, and Pengyang Wang. 2026. Shift-Resilient Diffusive Imputation for Variable Subset Forecasting. In *Proceedings of the ACM Web Conference 2026* (United Arab Emirates) *(WWW '26)*. Association for Computing Machinery, New York, NY, USA, 7366–7377. doi:10.1145/3774904.3792511

[29] Min Yang, Yang An, Jinliang Deng, Xiaoyu Li, Bin Xu, Ji Zhong, Xiankai Lu, and Yongshun Gong. 2025. CAN-ST: clustering adaptive normalization for spatio-temporal OOD learning. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence* (Montreal, Canada) *(IJCAI '25)*. Article 394, 9 pages. doi:10.24963/ijcai.2025/394

[30] Weiwei Ye, Songgaojun Deng, Qiaosha Zou, and Ning Gui. 2024. Frequency adaptive normalization for non-stationary time series forecasting. In *Proceedings of the 38th International Conference on Neural Information Processing Systems* (Vancouver, BC, Canada) *(NIPS '24)*. Curran Associates Inc., Red Hook, NY, USA, Article 985, 30 pages.

[31] Kun Yi, Qi Zhang, Wei Fan, Hui He, Liang Hu, Pengyang Wang, Ning An, Longbing Cao, and Zhendong Niu. 2023. FourierGNN: rethinking multivariate time series forecasting from a pure graph perspective. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) *(NIPS '23)*. Curran Associates Inc., Red Hook, NY, USA, Article 3050, 23 pages.

[32] Hsiang-Fu Yu, Nikhil Rao, and Inderjit S Dhillon. 2016. Temporal regularized matrix factorization for high-dimensional time series prediction. In *Advances in Neural Information Processing Systems*, Vol. 29.

[33] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. 2023. Are transformers effective for time series forecasting?. In *Proceedings of the Thirty-Seventh AAAI Conference on Artificial Intelligence and Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence and Thirteenth Symposium on Educational Advances in Artificial Intelligence (AAAI'23/IAAI'23/EAAI'23)*. AAAI Press, Article 1248, 8 pages. doi:10.1609/aaai.v37i9.26317

[34] Hanqing Zeng, Hongkuan Zhou, Ajitesh Srivastava, Rajgopal Kannan, and Viktor Prasanna. 2020. GraphSAINT: Graph Sampling Based Inductive Learning Method. In *International Conference on Learning Representations*. https://openreview.net/forum?id=BJe8pkHFwS

[35] Haoyu Zhang, Wentao Zhang, Hao Miao, Xinke Jiang, Yuchen Fang, and Yifan Zhang. 2025. STRAP: Spatio-Temporal Pattern Retrieval for Out-of-Distribution Generalization. *arXiv preprint arXiv:2505.19547* (2025).

[36] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. 2021. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. In *The Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021, Virtual Conference*, Vol. 35. AAAI Press, 11106–11115.

[37] Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. 2022. FEDformer: Frequency Enhanced Decomposed Transformer for Long-term Series Forecasting. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 162)*, Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato (Eds.). PMLR, 27268–27286. https://proceedings.mlr.press/v162/zhou22g.html

A Supplementary Proofs

A.1 Theoretical Analysis of Spatio-Temporal Shift Coupling

In this part, we provide a rigorous theoretical analysis to validate the inherent coupling between intra-series and inter-series distribution shifts. We demonstrate a bidirectional constraint mechanism where spatial correlations are mathematically bound by temporal variations, and conversely, temporal mean drifts are constrained by spatial fluctuations.

Setup and Notation. Consider a N-dimensional time series $X(\tau)$ and two distinct time windows W_1, W_2. For a variable pair (i, j) in window W_ℓ ($\ell \in \{1, 2\}$), let $\mu_\ell \in \mathbb{R}^2$ be the mean vector, $M_\ell \in \mathbb{R}^{2\times 2}$ the second moment matrix, and $\Sigma_\ell = M_\ell - \mu_\ell \mu_\ell^T$ the covariance matrix. We assume the variables satisfy variance lower bounds ($\mathrm{Var}(X) \geq v_{\min} > 0$) and structural upper bounds ($\|\mu_\ell\|_2 \leq M_{\max}$, $\|M_\ell\|_2 \leq M_{\max}$). Finally, the absolute Pearson correlation difference is formally denoted as $\Delta\rho_{ij} = |\rho_{ij}^{(1)} - \rho_{ij}^{(2)}|$.

Proposition A.1 (Upper Bound for Correlation Shift under Temporal Shift). *The pairwise Pearson correlation shift $\Delta\rho_{ij}$ is linearly bounded by the changes in first- and second-order moments:*

$$\Delta\rho_{ij} \leq K\left(\Delta M_{ij} + \Delta\mu_{ij}\right), \tag{24}$$

where $\Delta M_{ij} = \|M_1 - M_2\|_2$ and $\Delta\mu_{ij} = \|\mu_1 - \mu_2\|_2$. The constant K is distribution-free and depends on universal bounds $v_{\min}$ and $M_{\max}$.

Proof. First, we bound the covariance matrix difference using $\Sigma_\ell = M_\ell - \mu_\ell \mu_\ell^T$ and the triangle inequality:

$$\|\Sigma_1 - \Sigma_2\|_2 \leq \|M_1 - M_2\|_2 + \|\mu_1\mu_1^T - \mu_2\mu_2^T\|_2 \leq \Delta M_{ij} + 2M_{\max}\Delta\mu_{ij}. \tag{25}$$

Let $a = \sigma_{1,ij}, b = \sigma_{1,i}^2, c = \sigma_{1,j}^2$ (with primes denoting corresponding statistics in W_2). We can decompose the correlation shift as:

$$\Delta\rho_{ij} = \left|\frac{a}{\sqrt{bc}} - \frac{a'}{\sqrt{b'c'}}\right| \leq \frac{|a - a'|}{\sqrt{bc}} + |a'|\left|\frac{1}{\sqrt{bc}} - \frac{1}{\sqrt{b'c'}}\right|. \tag{26}$$

Since variances are bounded ($\sqrt{bc} \geq v_{\min}$), the first term is strictly bounded by $\|\Sigma_1 - \Sigma_2\|_2 / v_{\min}$. For the second term, we bound $|a'| \leq \|\Sigma_2\|_2 \leq M_{\max} + M_{\max}^2 := B_{\max}$. Through algebraic simplification, the reciprocal difference is bounded by $\frac{B_{\max}}{v_{\min}^3}\|\Sigma_1 - \Sigma_2\|_2$. Combining these derived components yields:

$$\Delta\rho_{ij} \leq \left(\frac{1}{v_{\min}} + \frac{B_{\max}^2}{v_{\min}^3}\right)\|\Sigma_1 - \Sigma_2\|_2. \tag{27}$$

Substituting Eq. (25) into this inequality gives $\Delta\rho_{ij} \leq K(\Delta M_{ij} + \Delta\mu_{ij})$, where $K = \left(\frac{1}{v_{\min}} + \frac{B_{\max}^2}{v_{\min}^3}\right)\max\{1, 2M_{\max}\}$, thereby successfully completing the proof. □

Proposition A.2 (Temporal Mean Shift Bounded by Second-Order Shifts). *Assume the mean vectors are non-degenerate ($\|\mu_\ell\|_2 \geq m_{\min} > 0$) and satisfy a mild directional alignment condition: $\mu_1^T\mu_2 \geq c\|\mu_1\|_2\|\mu_2\|_2$ for some $c > -1$. Then, the temporal mean shift is upper bounded by:*

$$\Delta\mu_{ij} \leq K'(\Delta M_{ij} + \Delta\Sigma_{ij}), \tag{28}$$

where $\Delta\Sigma_{ij} = \|\Sigma_1 - \Sigma_2\|_2$, and $K' = 2/(m_{\min}\sqrt{2(1+c)})$.

Proof. Rearranging the covariance definition gives $\mu_1\mu_1^T - \mu_2\mu_2^T = (M_1 - M_2) - (\Sigma_1 - \Sigma_2)$. Taking spectral norms and applying the triangle inequality yields:

$$\|\mu_1\mu_1^T - \mu_2\mu_2^T\|_2 \leq \|M_1 - M_2\|_2 + \|\Sigma_1 - \Sigma_2\|_2 = \Delta M_{ij} + \Delta\Sigma_{ij}. \tag{29}$$

Using the symmetric decomposition identity, $\mu_1\mu_1^T - \mu_2\mu_2^T = \frac{1}{2}(\mu_1 - \mu_2)(\mu_1 + \mu_2)^T + \frac{1}{2}(\mu_1 + \mu_2)(\mu_1 - \mu_2)^T$. The spectral norm of this difference is lower-bounded by $\frac{1}{2}\|\mu_1 + \mu_2\|_2\Delta\mu_{ij}$. Under the alignment assumption and magnitude bounds, we have:

$$\|\mu_1 + \mu_2\|_2 = \sqrt{\|\mu_1\|_2^2 + \|\mu_2\|_2^2 + 2\mu_1^T\mu_2} \geq m_{\min}\sqrt{2(1+c)}. \tag{30}$$

Substituting this lower bound into Eq. (29) gives $\frac{1}{2}m_{\min}\sqrt{2(1+c)}\Delta\mu_{ij} \leq \Delta M_{ij} + \Delta\Sigma_{ij}$. Rearranging for $\Delta\mu_{ij}$ yields the desired bound. □

A.2 Theoretical Analysis of Invariance of Pearson Correlation under Instance Normalization

In this section, we provide a rigorous mathematical proof demonstrating that the temporal instance normalization (RevIN) operation preserves the Pearson Correlation Coefficient (PCC) between any pair of variables. This property ensures that the spatial topology extracted after normalization remains faithful to the intrinsic inter-variable dependencies of the raw data.

Proposition A.3. *Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^L$ be two time series vectors within a look-back window of length L. Let $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ denote the sequences obtained by applying instance normalization to $\mathbf{x}$ and $\mathbf{y}$, respectively. The Pearson Correlation Coefficient is invariant under this transformation, i.e., $PCC(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = PCC(\mathbf{x}, \mathbf{y})$.*

Proof. By definition, instance normalization applies the transformation $\hat{x}_t = \frac{x_t - \mu_x}{\sigma_x}$ to a sequence $\mathbf{x}$ (where μ_x and σ_x are the sample mean and standard deviation). This yields a normalized sequence $\hat{\mathbf{x}}$ with zero mean ($\mu_{\hat{x}} = 0$) and unit variance ($\sigma_{\hat{x}} = 1$).

Leveraging these statistical properties, the Pearson Correlation Coefficient between the normalized sequences $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ can be directly simplified and expanded as follows:

$$\begin{aligned}
\mathrm{PCC}(\hat{\mathbf{x}}, \hat{\mathbf{y}}) &= \frac{\mathrm{Cov}(\hat{\mathbf{x}}, \hat{\mathbf{y}})}{\sigma_{\hat{x}}\sigma_{\hat{y}}} = \frac{1}{L}\sum_{t=1}^{L}(\hat{x}_t - 0)(\hat{y}_t - 0) \\
&= \frac{1}{L}\sum_{t=1}^{L}\left(\frac{x_t - \mu_x}{\sigma_x}\right)\left(\frac{y_t - \mu_y}{\sigma_y}\right) \\
&= \frac{1}{\sigma_x\sigma_y}\left[\frac{1}{L}\sum_{t=1}^{L}(x_t - \mu_x)(y_t - \mu_y)\right] \\
&= \frac{\mathrm{Cov}(\mathbf{x}, \mathbf{y})}{\sigma_x\sigma_y} \equiv \mathrm{PCC}(\mathbf{x}, \mathbf{y}).
\end{aligned} \tag{31}$$

This concludes the proof. As instance normalization acts as a strictly positive affine transformation, it inherently preserves the linear correlation structure between variables. □

A.3 Theoretical Analysis of Low-Rank Manifold Purification

In this section, we provide a theoretical justification for the Low-rank Manifold Spatial Decoupling (LMSD) module. By forcing the input variant pattern $\mathbf{X} \in \mathbb{R}^{N\times L}$ through a low-dimensional bottleneck ($K \ll N$), LMSD acts as a structural filter. Recall from Section 4.2.1 that, using the community assignment matrix $\mathbf{S} \in \mathbb{R}^{N\times K}$, the operator follows a pooling-broadcasting scheme $\hat{\mathbf{X}} = \mathbf{S}\,\mathrm{Norm}_2(\mathbf{S}^\top\mathbf{X})$, where $\mathrm{Norm}_2(\cdot)$ normalizes each community representative by its effective volume $M_k = \sum_{j=1}^{N} S_{jk}$. For analytical convenience, we equivalently express it in the linear form:

$$\hat{\mathbf{X}} = \mathcal{P}_{\mathcal{M}}(\mathbf{X}) = \mathbf{S}\,\mathbf{\Lambda}\,\mathbf{S}^\top\mathbf{X}, \qquad \mathbf{\Lambda} = \mathrm{diag}(M_1^{-1}, \ldots, M_K^{-1}). \tag{32}$$

This imposes a hard rank constraint $\mathrm{rank}(\hat{\mathbf{X}}) \le K$, nullifying high-rank components orthogonal to the manifold span $\mathbf{S}$.

Proposition A.4 (Dual Property of Rank-Constrained Purification). *The operator $\mathcal{P}_{\mathcal{M}}$ attenuates localized stochastic perturbations (high-rank noise) in proportion to the community size, while preserving synchronized systematic shifts (low-rank signals).*

Proof. **Noise Attenuation:** Consider a localized perturbation $\mathbf{n} \in \mathbb{R}^N$ of magnitude η at node i, assuming a hard assignment approximation to community k ($S_{ik} \approx 1, S_{ip\neq k} \approx 0$). Projecting and reconstructing this noise yields the output at node i:

$$\hat{n}_i = (\mathbf{S}\mathbf{\Lambda}\mathbf{S}^\top\mathbf{n})_i = \eta \sum_{p=1}^{K} \frac{S_{ip}^2}{M_p} \approx \frac{\eta}{M_k}. \tag{33}$$

Since $M_k \gg 1$, the noise magnitude is drastically attenuated. The vast majority of the noise energy lies in the kernel of the rank-K projection and is discarded.

Signal Preservation (Low-Rank Retention): Conversely, consider a low-rank coherent shift $\mathbf{L} \in \mathbb{R}^N$ where all nodes in community k exhibit a synchronized change of magnitude Δ. The latent representation accurately aggregates this: $c_k = \frac{1}{M_k}\sum_{j\in C_k}\Delta = \Delta$. The reconstructed signal at any node $i \in C_k$ is then $\hat{L}_i \approx S_{ik}c_k \approx \Delta$. Because $\mathbf{L}$ is inherently low-rank and aligned with the basis $\mathbf{S}$, it acts as an approximate eigenvector of $\mathcal{P}_{\mathcal{M}}$ with an eigenvalue of 1, passing through without information loss. □

Consequently, LMSD achieves a significant Signal-to-Noise Ratio (SNR) improvement, effectively separating signals based on their manifold alignment:

$$\frac{\|\mathcal{P}_{\mathcal{M}}(\mathbf{N})\|_F}{\|\mathbf{N}\|_F} \approx O\left(\frac{1}{\sqrt{M_k}}\right) \quad \ll \quad \frac{\|\mathcal{P}_{\mathcal{M}}(\mathbf{L})\|_F}{\|\mathbf{L}\|_F} \approx 1. \tag{34}$$

B Implementation Details

B.1 Theoretical Rationality of Differentiable Community Assignment

To ensure the soft assignment matrix $\mathbf{S}_t \in \mathbb{R}^{N\times K}$ learned by the LMSD module captures a physically meaningful topology, we explicitly impose a compound structural regularization: $\mathcal{L}_{\text{struct}} = \lambda_1(\mathcal{L}_{\text{mincut}} + \mathcal{L}_{\text{orth}})$. This formulation mathematically enforces both intra-community cohesion and inter-community separation.

MinCut Loss (Maximizing Cohesion): This term relaxes the standard Normalized Cut problem to group variables exhibiting strong historical correlations:

$$\mathcal{L}_{\text{mincut}} = -\frac{\mathrm{Tr}(\mathbf{S}_t^\top\mathbf{M}^t\mathbf{S}_t)}{\mathrm{Tr}(\mathbf{S}_t^\top\mathbf{D}\mathbf{S}_t)}. \tag{35}$$

The numerator, representing the association strength, expands to $\sum_{k,i,j}(\mathbf{S}_t)_{ik}(\mathbf{M}^t)_{ij}(\mathbf{S}_t)_{jk}$. This sum is maximized *if and only if* strongly correlated variable pairs (large stable memory $(\mathbf{M}^t)_{ij}$) are simultaneously assigned high probabilities for the same community k. The denominator normalizes by community volume, driving dense intra-cluster connections while averting trivial graph cuts.

Orthogonality Loss (Separation and Balance): This enforces distinctiveness and load balance among the learned communities:

$$\mathcal{L}_{\text{orth}} = \left\| \frac{\mathbf{S}_t^\top\mathbf{S}_t}{\|\mathbf{S}_t^\top\mathbf{S}_t\|_F} - \frac{\mathbf{I}_K}{\sqrt{K}} \right\|_F^2. \tag{36}$$

Let $\mathbf{G} = \mathbf{S}_t^\top\mathbf{S}_t \in \mathbb{R}^{K\times K}$ denote the community overlap matrix. Aligning $\mathbf{G}$ with a scaled identity matrix enforces two key properties: (1) *Orthogonality:* Minimizing the off-diagonal elements ($\mathbf{G}_{kp} = \sum_i(\mathbf{S}_t)_{ik}(\mathbf{S}_t)_{ip} \to 0$) forces mutually exclusive assignments, ensuring a single variable does not exhibit high probabilities for multiple communities. (2) *Balance:* Forcing the diagonal elements ($\mathbf{G}_{kk}$, reflecting the effective size of community k) to be uniform acts as an entropy regularization. This ensures balanced partition sizes across the K low-rank manifolds and prevents mode collapse where one community dominates the representation.

B.2 Details of Stability Bounding Mechanism

In the Denormalization module, directly utilizing raw neural network outputs as affine transformation parameters can cause severe training instability, particularly gradient explosion during the subsequent variance exponentiation step. To mitigate these issues, we design bounding functions $\mathcal{B}_\mu(\cdot)$ and $\mathcal{B}_r(\cdot)$ to strictly constrain the estimated shift adjustments ($\Delta\boldsymbol{\mu}_t$ and $\mathbf{r}_t$) within a plausible range.

Mean Shift Bounding ($\mathcal{B}_\mu$). To ensure the predicted mean shift is proportional to the inherent volatility of the series, we scale the localized shift by the historical standard deviation $\boldsymbol{\sigma}_t$:

$$\mathcal{B}_\mu(\Delta\boldsymbol{\mu}_t) = g_\mu \cdot \boldsymbol{\sigma}_t \odot \tanh(\Delta\boldsymbol{\mu}_t), \tag{37}$$

where $\tanh(\cdot)$ compresses the raw prediction into $(-1, 1)$, and g_μ is a learnable scalar gain that adaptively regulates the correction amplitude (initialized to a small positive value). This formulation allows the model to learn relative shifts while preventing the predicted mean from drifting arbitrarily far from the input distribution.

Scale Factor Bounding ($\mathcal{B}_r$). For the scale adjustment, we employ a hard clipping mechanism to prevent numerical instability in the exponential domain:

$$\mathcal{B}_r(\mathbf{r}_t) = \mathrm{clip}\big(g_s \cdot \tanh(\mathbf{r}_t), -c, c\big), \tag{38}$$

where g_s is a learnable gain and c is a pre-defined threshold. The $\mathrm{clip}(\cdot, -c, c)$ operation strictly restricts the log-space adjustment to the interval $[-c, c]$, ensuring that the recovered standard deviation $\boldsymbol{\sigma}_{\text{final},t} = \boldsymbol{\sigma}_t \odot \exp(\mathcal{B}_r(\mathbf{r}_t))$ remains finite and numerically stable.

Final Restoration. Combining these bounded components, the final restoration formula is:

$$\hat{\mathbf{Y}}_{t+1:t+H} = \hat{\mathbf{Y}}_{\text{inv},t} \odot (\boldsymbol{\sigma}_t \odot \exp(\mathcal{B}_r(\mathbf{r}_t))) + \big(\boldsymbol{\mu}_t + \mathcal{B}_\mu(\Delta\boldsymbol{\mu}_t)\big). \tag{39}$$